RAJALAKSHMI ENGINEERING COLLEGE

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R2

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|---|---|
| **Started on** | Thursday, 8 October 2026, 10:56 AM |
| **State** | Finished |
| **Completed on** | Thursday, 8 October 2026, 10:57 AM |
| **Time taken** | 1 min 42 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, i;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    int duplicate = -1;
    for (i = 0; i < n; i++) {
        int index = a[i] < 0 ? -a[i] : a[i];
        if (a[index] < 0) {
            duplicate = index;
            break;
        }
        a[index] = -a[index];
    }
    printf("%d\n", duplicate);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 11<br>10 9 7 6 5 1 2 3 8 4 7 | 7 | 7 | ✔ |
| ✔ | 5<br>1 2 3 4 4 | 4 | 4 | ✔ |
| ✔ | 5<br>1 1 2 3 4 | 1 | 1 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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